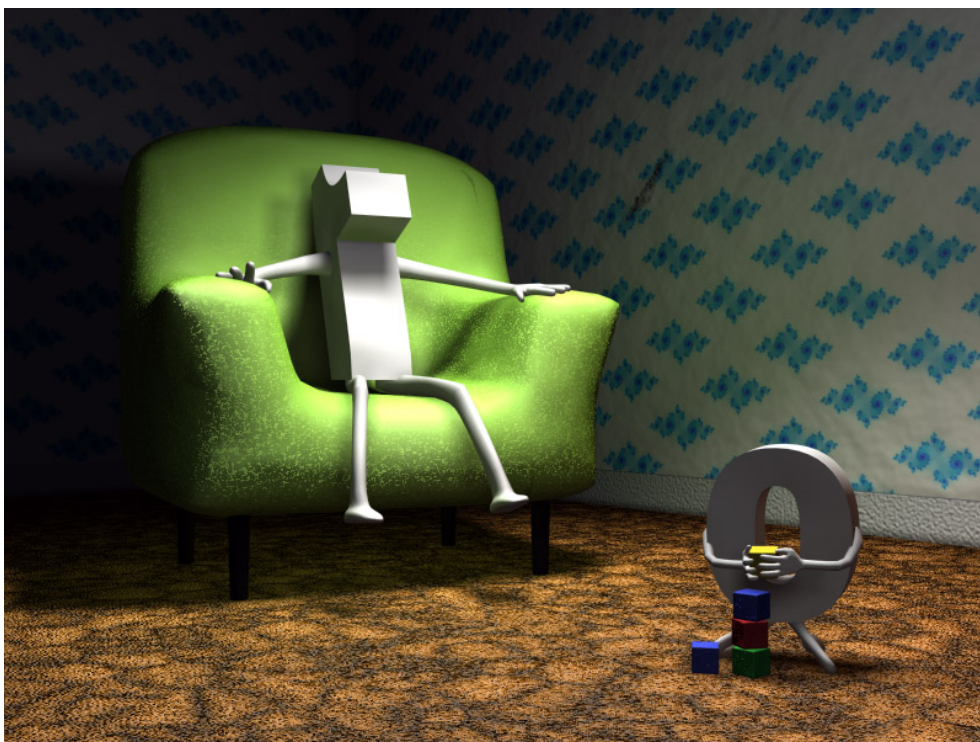


Learning how to prove Workshop Week 3

Integers



Solving equations in $\mathbb{Z}_n$. The Chinese Remainder Theorem

1. Let p be a prime.
 - (a) Prove that the only elements in $\mathbb{Z}_p$ that are self-inverses are $\bar{1}$ and $\overline{p-1}$.
 - (b) **Prove Wilson's Theorem:** $(p-1)! \equiv -1 \pmod{p}$

2. Use the Chinese Remainder Theorem to solve the following simultaneous congruence:

$$3x \equiv 2 \pmod{5} \quad \text{and} \quad 2x \equiv 1 \pmod{3}.$$